

Terraform Task-2

Task Description:

Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

Step 1 : Configured aws cli in terraform server.

```
ubuntu@ip-172-31-40-191:~$ aws configure
AWS Access Key ID [None]: AKIA4F35FI6UL3H0DB60
AWS Secret Access Key [None]: /5Hoewbtd3FtkUR/vLbMkmICmkM622qckkwhYVbJ
Default region name [None]: us-west-2
Default output format [None]:
```

Step 2 - Created a terraform script to launch instances in N. Verginia and Mumbai

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  alias = "us"
```

```

    region = "us-east-1"
}

provider "aws" {

    alias = "in"

    region = "ap-south-1"
}

data "aws_ami" "al2023_us" {

    provider   = aws.us

    most_recent = true

    owners     = ["amazon"]

    filter {

        name = "name"

        values = ["al2023-ami-*-x86_64"]
    }
}

data "aws_ami" "al2023_in" {

    provider   = aws.in

    most_recent = true

    owners     = ["amazon"]

    filter {

        name = "name"

        values = ["al2023-ami-*-x86_64"]
    }
}

resource "aws_security_group" "web_us" {

    provider = aws.us

```

```
name    = "nginx-sg-us"

ingress {

  from_port = 22

  to_port   = 22

  protocol  = "tcp"

  cidr_blocks = ["0.0.0.0/0"]

}

ingress {

  from_port = 80

  to_port   = 80

  protocol  = "tcp"

  cidr_blocks = ["0.0.0.0/0"]

}

egress {

  from_port = 0

  to_port   = 0

  protocol  = "-1"

  cidr_blocks = ["0.0.0.0/0"]

}

}

resource "aws_security_group" "web_in" {

  provider = aws.in

  name     = "nginx-sg-in"

  ingress {

    from_port = 22

    to_port   = 22
```

```
protocol = "tcp"

cidr_blocks = ["0.0.0.0/0"]

}

ingress {

    from_port = 80

    to_port = 80

    protocol = "tcp"

    cidr_blocks = ["0.0.0.0/0"]

}

egress {

    from_port = 0

    to_port = 0

    protocol = "-1"

    cidr_blocks = ["0.0.0.0/0"]

}

}

resource "aws_instance" "nginx_us" {

    provider = aws.us

    ami = data.aws_ami.al2023_us.id

    instance_type = "t3.micro"

    security_groups = [

        aws_security_group.web_us.name

    ]

    user_data = <<-EOF

    #!/bin/bash

    dnf update -y
```

```
dnf install -y nginx
```

```
systemctl enable nginx
```

```
systemctl start nginx
```

```
EOF
```

```
tags = {
```

```
    Name = "nginx-us-east-1"
```

```
}
```

```
}
```

```
resource "aws_instance" "nginx_in" {
```

```
    provider    = aws.in
```

```
    ami         = data.aws_ami.al2023_in.id
```

```
    instance_type = "t3.micro"
```

```
    security_groups = [
```

```
        aws_security_group.web_in.name
```

```
    ]
```

```
    user_data = <<-EOF
```

```
        #!/bin/bash
```

```
        dnf update -y
```

```
        dnf install -y nginx
```

```
        systemctl enable nginx
```

```
        systemctl start nginx
```

```
EOF
```

```
tags = {
```

```
    Name = "nginx-ap-south-1"
```

```
}
```

```
}
```

Step 3 - Executed the script

terraform init

```
ubuntu@ip-172-31-40-191:~/project2$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!
```

terraform apply

```
ubuntu@ip-172-31-40-191:~/project2$ terraform apply
data.aws_ami.al2023_us: Reading...
data.aws_ami.al2023_in: Reading...
data.aws_ami.al2023_us: Read complete after 1s [id=ami-02f058bc6b3ee6fcc]
data.aws_ami.al2023_in: Read complete after 2s [id=ami-02a8bcaaf6d87b759]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:
```

Instances started running in two regions

Instance summary for i-01077ae44608eb7bd (nginx-us-east-1) [Info](#)

Updated less than a minute ago

Instance ID i-01077ae44608eb7bd	Public IPv4 address 52.91.170.104 open address	Private IPv4 addresses 172.31.21.100
IPv6 address -	Instance state Running	Public DNS ec2-52-91-170-104.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-21-100.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-21-100.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t5.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 52.91.170.104 [Public IP]	VPC ID vpc-03e1852553eb9dd75	Auto Scaling Group name -
IAM role -	Subnet ID subnet-02e8cc4a115b24ff6	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:837244241832:instance/i-01077ae44608eb7bd	
Operator		

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Instance summary for i-0348ead604e4d5687 (nginx-ap-south-1) [Info](#)

🔄 Refreshing instance data

Instance ID

[🔗](#) i-0348ead604e4d5687

IPv6 address

–

Hostname type

IP name: ip-172-31-44-115.ap-south-1.compute.internal

Answer private resource DNS name

–

Auto-assigned IP address

🔄

IAM role

–

IMDSv2

Required

Operator

–

Public IPv4 address

[🔗](#) 13.232.130.244 | [open address 🔗](#)

Instance state

🟢 Running

Private IP DNS name (IPv4 only)

[🔗](#) ip-172-31-44-115.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

[🔗](#) vpc-00f1b106cd708742 | [🔗](#)

Subnet ID

[🔗](#) subnet-003ad8b0f91fd320d | [🔗](#)

Instance ARN

[🔗](#) arn:aws:ec2:ap-south-1:837244241832:instance/i-0348ead604e4d5687

Private IPv4 addresses

[🔗](#) 172.31.44.115

Public DNS

[🔗](#) ec2-13-232-130-244.ap-south-1.compute.amazonaws.com | [open address 🔗](#)

Elastic IP addresses

🔄

AWS Compute Optimizer finding

[🔗](#) Opt-in to AWS Compute Optimizer for recommendations. | [Learn more 🔗](#)

Auto Scaling Group name

–

Managed

false

[Connect](#)[Instance state ▼](#)[Actions ▼](#)

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